

Daniele Famà

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Seeking a 6 month End-of-Studies Internship / Master's Thesis starting April under a French convention de stage

EDUCATION

Télécom Paris, Institut Polytechnique de Paris

2025 – 2027

M.Sc. in Engineering – Double Degree Program

Paris, France

- Year 1 tracks: Distributed Software Systems, Applied Algebra (Cryptography, Quantum Information)
- Year 2 specialization: Artificial Intelligence

Politecnico di Torino

2024 – 2025

M.Sc. in Data Science and Engineering – GPA: 29/30 (Double Degree)

Turin, Italy

University of Catania

2021 – 2024

B.Sc. in Computer Engineering – 110/110, summa cum laude (Highest Honors)

Catania, Italy

- Thesis: Design of a Third-Order Tow-Thomas Butterworth Filter for Audio-Band Applications

EXPERIENCE

LTCI, Institut Polytechnique de Paris

2026

Research Intern – Energy-Aware Neural Architecture Search

Paris, France

- Engineered an **energy-aware NAS framework** (Python, PyTorch) to search for optimal sub-networks via differentiable structured pruning and empirical hardware regularizers.
- Physically validated **up to 60% real hardware energy reduction** on ResNet-18 deployed on edge devices (Raspberry Pi 5, NVIDIA Jetson) with **zero accuracy loss**.
- Automated power profiling across **2,000+ layer configurations**, capturing physical sensor telemetry on Jetson and Raspberry Pi boards.

PROJECTS

ARDA VSLNet: Efficient Video-Language Localization on Ego4D

2025

Multimodal Deep Learning & Optimization

Politecnico di Torino

- Enhanced natural-language video localization on Ego4D, outperforming baselines by **19.73%**, by integrating FiLM conditioning into the VSLNet architecture.
- Compressed video grounding models while retaining accuracy, cutting model size by **88.47%** and compute by **80.6% GFLOPs**, by applying block-wise knowledge distillation.

Distributed MapReduce on Lab Machines

2026

Distributed Systems & Concurrency

Télécom Paris

- Achieved a **5.45x scaling speedup across 64 unprivileged lab nodes** on Common Crawl by architecting a disk-backed streaming TSV shuffle in **Go** that eliminated memory bottlenecks.
- Guaranteed zero data loss with **sub-10s recovery** during worker crashes by implementing an epoch-tracked slot state machine activating cold spares.
- Outperformed an 8-node Kafka baseline by **38x** (0.266s vs 10.10s) across 4 analytical jobs by engineering modular Go worker interfaces and automated harnesses.

Speech-Based Age Estimation

2025

Audio Machine Learning & Data Pipelines

Politecnico di Torino

- Secured **1st place out of 100+ teams** on the course public leaderboard by engineering an end-to-end ensemble regression pipeline combining CatBoost and Random Forest.
- Maximized model robustness on out-of-distribution audio by extracting hybrid Mel-spectrogram, spectral, and statistical features paired with tuned dimensionality reduction.

SKILLS

Programming : Python, C/C++, Go, MATLAB

ML and Data : PyTorch, TensorFlow, scikit-learn, Spark

Platforms : Linux, Git, NVIDIA Jetson, Raspberry Pi

Languages : Italian (Native), English (C1, Advanced), French (B1, Intermediate)